

KlangTresor

Your Suno music. At home.

github.com/CasparDavi/klangtresor

Suno makes the songs. KlangTresor keeps them, plays them, measures them and shows them, on every screen in your home.



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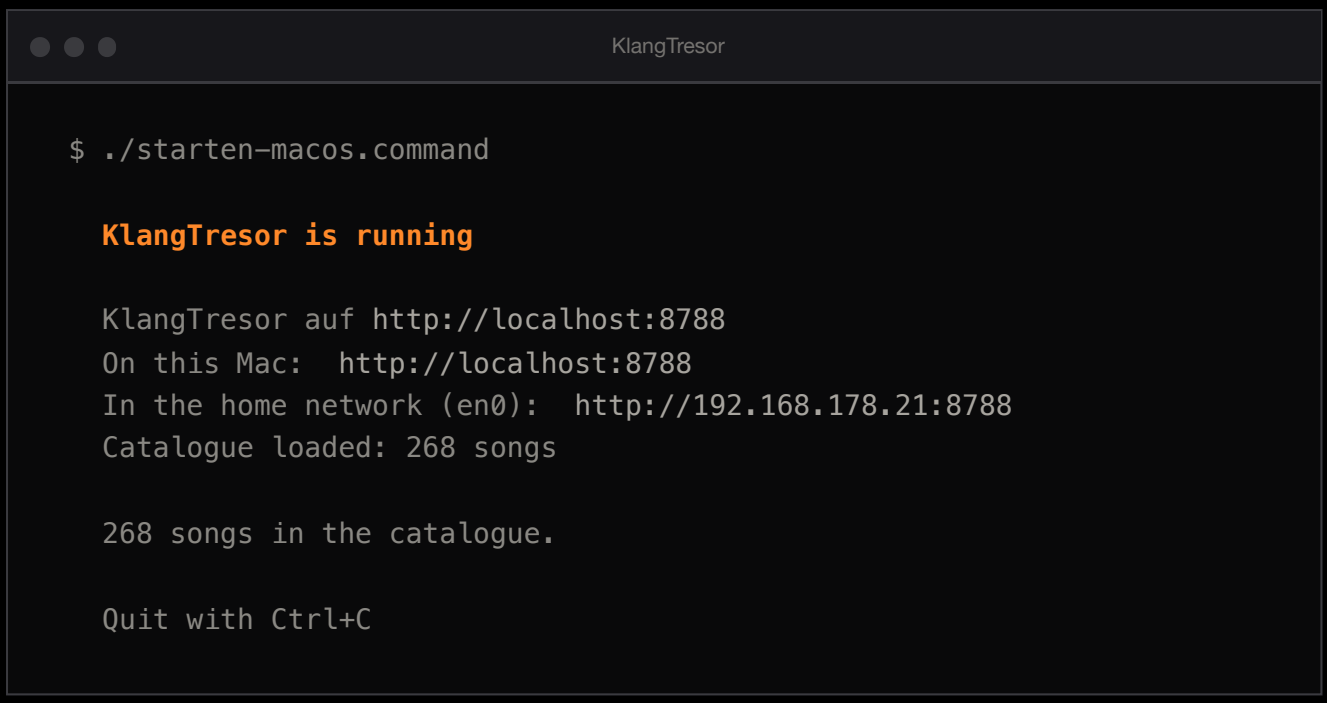
01 SETUP

Runs at home. Plays anywhere on your Wi-Fi.

One computer keeps the vault. Every screen in your home network opens it.

01.1
Where it runs
p. 04

01.2
Who reads it
p. 04



01.1
WHERE IT RUNS

One computer at home.

macOS, Windows, Linux or Docker. Set up with a double click (on Linux, one command). Your archive is a folder on its disk.

- **Needs Node.js and ffmpeg.** The setup brings Node.js along; it fetches ffmpeg itself on Windows, on Mac and Linux one command does it.
- **Space:** about 1.6 GB for the program and its models, plus 3.1 GB for Whisper; then about 160 MB per song with everything that belongs to it: WAV, instrument tracks, analysis, covers (measured on the author's 343 songs).
- **The computer is the server.** While its window is open, your vault is reachable; close it and the vault sleeps.

01.2
WHO READS IT
Fig. p. 03

Every screen with a browser.

Laptop, iPhone, Android, iPad, tablet. Anything that runs a web browser in your home network. Nothing to install.

- **Scan a QR code** on the computer and the phone opens your vault.
- **On the home screen** like an app; on the iPhone without the address bar.
- **Home network only.** From outside only if you open it on your router. Inside there is no password: whoever is on your network and knows the address can listen.



02 COLLECT

Suno fills it. You keep it.

One click on the bookmark at Suno and on the red button, and the vault fetches what is new. If Suno changes, or your subscription ends, everything is already at home.

02.1
Morning run
p. 06

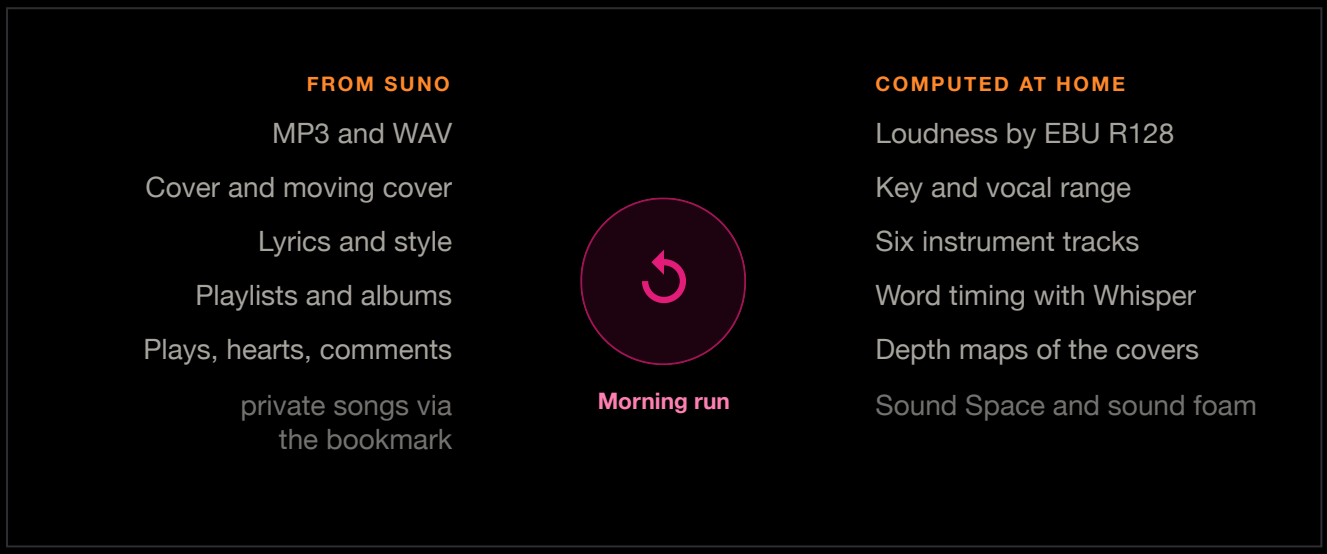
02.2
What travels
p. 06

02.1
MORNING RUN
Fig. p. 05

Everything you made. In full.

Covers, moving covers, lyrics, playlists, comments and hearts. The audio, MP3 and lossless WAV, it takes straight from your downloads; whatever you have not downloaded yet, you unlock at Suno.

- **Private songs too**, as long as they sit in one of your own playlists.
- **Your Suno login stays in your browser.** A bookmark does the fetching; KlangTresor never sees a password.
- **Only new things travel.** What is already at home is not fetched again.



02.2
WHAT TRAVELS

From Suno into your vault.

Suno delivers the songs. Everything that can be measured is computed at home.

- **The red button** at the top of KlangTresor starts the morning run by hand.

!

Good to know

- KlangTresor does not make music. Songs are made on Suno; KlangTresor keeps and shows them.
- It only fetches your own songs, with your own Suno login.
- When Suno changes its website, the morning run may need an update. Everything already at home stays playable.

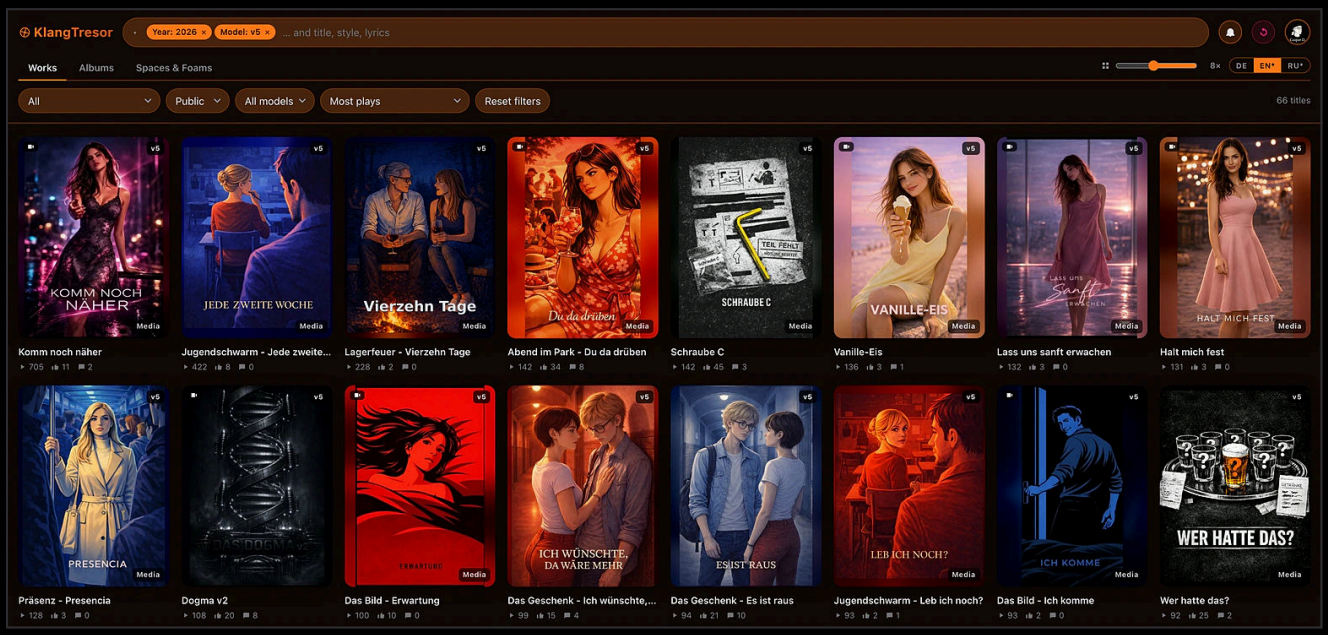
03

OVERVIEW

Everything at a glance.

Your whole work as a wall of covers. Filter, sort, and one click plays.

03.1 Filter, sort, listen p. 07
03.2 What you can filter by p. 07



03.1

FILTER, SORT, LISTEN

Find any song in seconds.

Tags in the search field, choices below, the player always at hand.

- **Tile size** from a few large covers to ten in a row.
- **A click on a cover** plays the song; the stage is one more click away.

IN THE SEARCH FIELD	Title · Style · Lyrics
AS A TAG	Year · Month · Since · Until · Model · Origin · Commented by · Missing: WAV, timestamps, analysis, comments, lyrics
CHOICES	With / without motion artwork · Public / private / both · Model
SORT	Newest · Oldest · Most plays · Most hearts · Most comments · Title A–Z · Longest · Fastest · Loudest · Most dynamic · Widest loudness range · Widest stereo

03.2

WHAT YOU CAN FILTER BY

Narrow it down.

Everything the vault knows about a song can become a filter.

- **Tags combine:** each one narrows the wall further.
- **One click on ×** removes a tag again.

04

LISTEN

A stage for every song.

Full screen, cover or moving image, and the lyrics follow the voice word by word.

04.1 Karaoke p. 09
04.2 Visualisations p. 09



04.1
KARAOKE
Fig. p. 08

Sing along. Word by word.

Word timing is computed on your own computer, for German, English and Russian songs.

- **Whisper runs locally.** Your lyrics never leave the house.
- **Readable from the sofa:** the current line large, the next one announced, the controls fade away.
- **Cover or moving cover** behind the text, and your own effect clips.



04.2
VISUALISATIONS

Light that listens.

Seven of its own, plus Butterchurn.

333 MilkDrop presets

- **Seven visualisations** built for KlangTresor.
- **Butterchurn** brings the classic MilkDrop look with 333 presets.

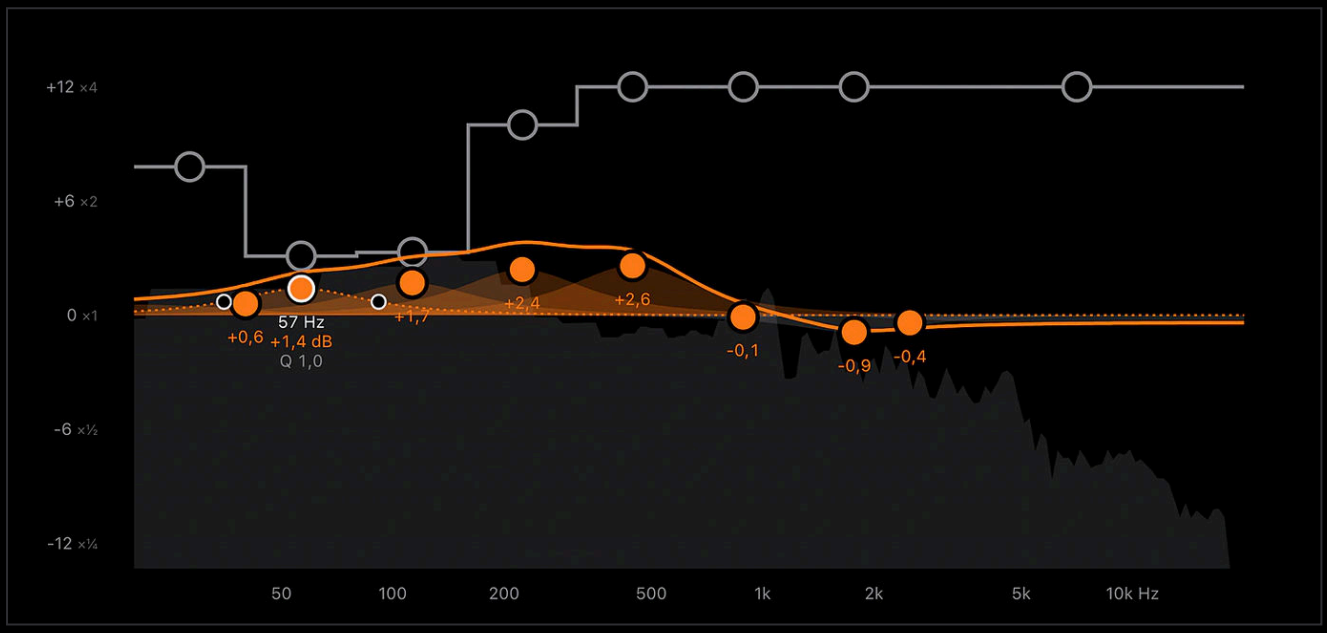
05

SOUND STUDIO

A real studio. In your browser.

Equaliser, bells, compressor, boombox and room calibration. One chain, in the order an engineer would build it.

05.1 8-band equaliser p. 10
05.2 Bell frame p. 10
05.3 Compressor p. 11
05.4 Boombox p. 11
05.5 Calibration p. 11



05.1

8-BAND EQUALISER

Eight bands. Your sound.

Start neutral, or let a song match the rest of your collection, its style group, its album or a reference song.

- **Every band shows its headroom.** The song itself decides how much a band can be raised.
- **Matching** to collection, style group, album or reference is computed from the measured spectra.
- **On the phone** as eight sliders with presets.

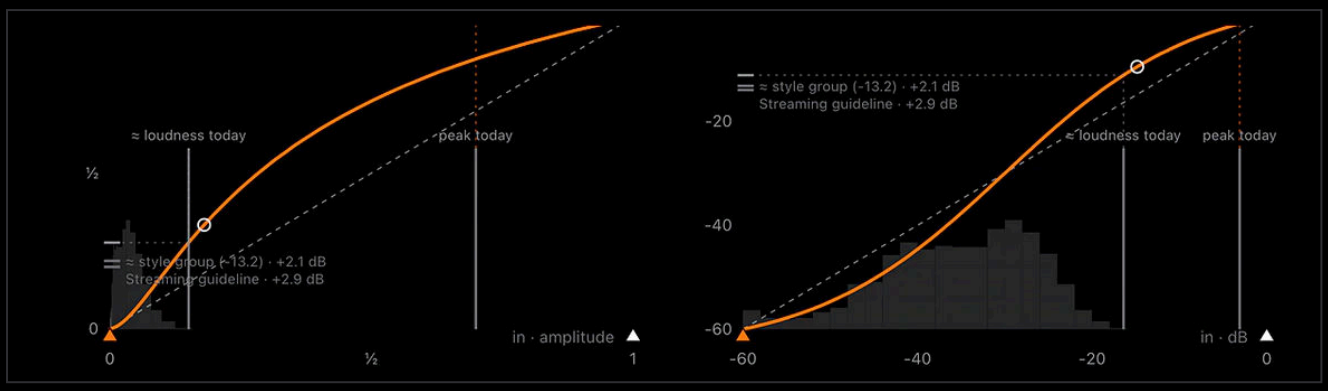
05.2

BELL FRAME

Bells you can grab.

Six bell filters, free in frequency, height and width, plus two shelves.

- **Drag a bell** to move it; the handles on its shoulders set its width.
- **Single or stacked:** see each bell on its own, or how they add up.
- **A notch** removes a single annoying frequency.

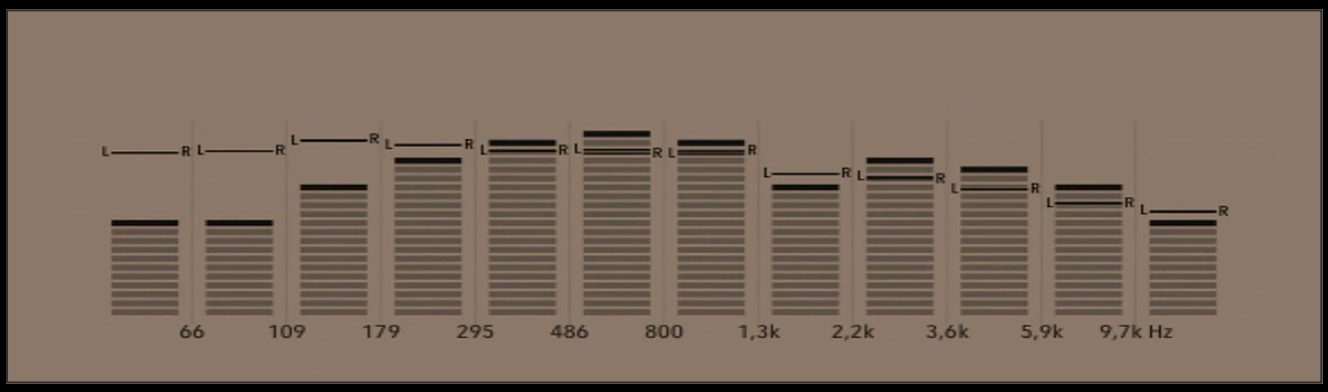


05.3
COMPRESSOR

Compressor. Or a gradation curve.

The classic knee with threshold, ratio and limiter, or a curve like in a photo editor: quiet point, loud point, gamma.

- **Two modes, one stage:** compressor or gradation compressor, never both at once.
- **The curve is drawn live**, with the gain reduction as it happens.
- **Gamma or sigmoid** with a single slider, for those who think in pictures.



05.4
BOOMBOX

One switch for everything.

Loudness, more bass, more treble, clear voice, night mode. Plus MEGA BASS, reverb, echo and stereo width.

- **Night mode** turns down the lows that travel through walls.
- **Tempo** faster or slower, with or without keeping the pitch.
- **A clip light** warns when everything together gets too loud.

05.5
CALIBRATION

Tuned to your room.

The microphone of your computer measures your speakers where you sit. The spectrum shows the whole chain: what Suno delivered, and what your settings make of it.

- **Measured at the listening position**, not in a lab.
- **Before and after** side by side in one spectrum.

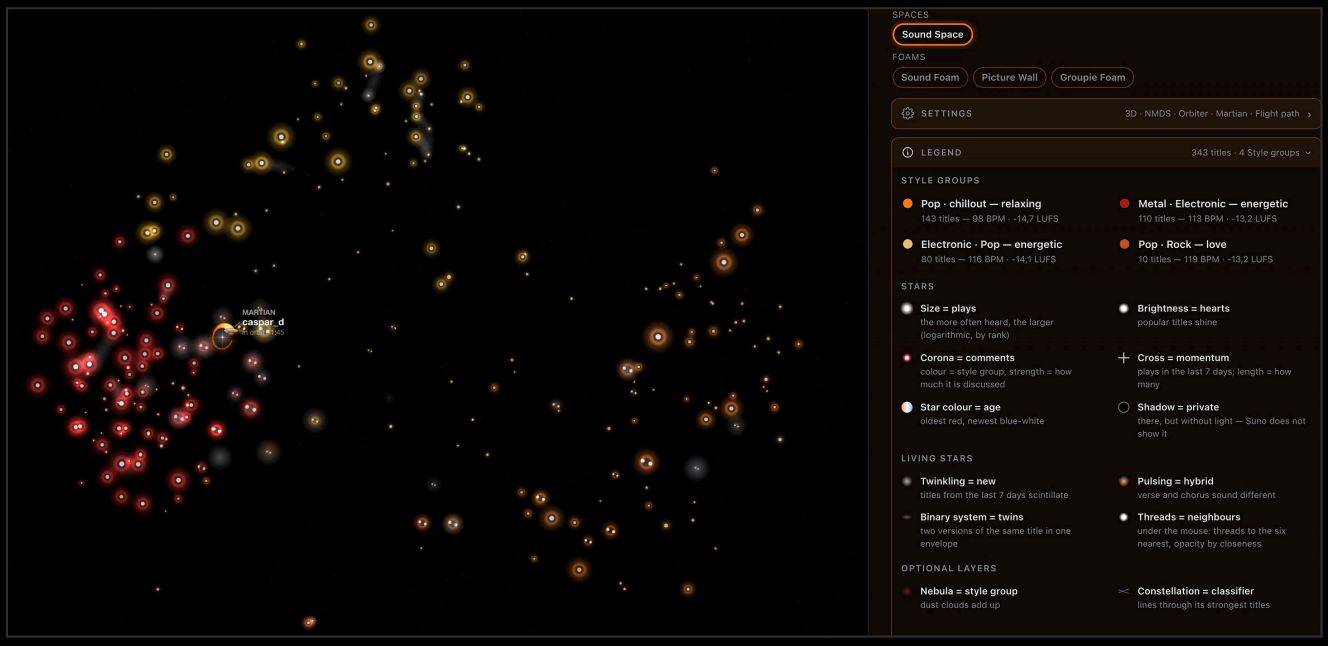
06

UNDERSTAND

Take a look inside your songs.

Every song measured. The whole collection as a sky of stars.

06.1 Sound Space p. 12
06.2 Analysis p. 13
06.3 What is measured p. 13

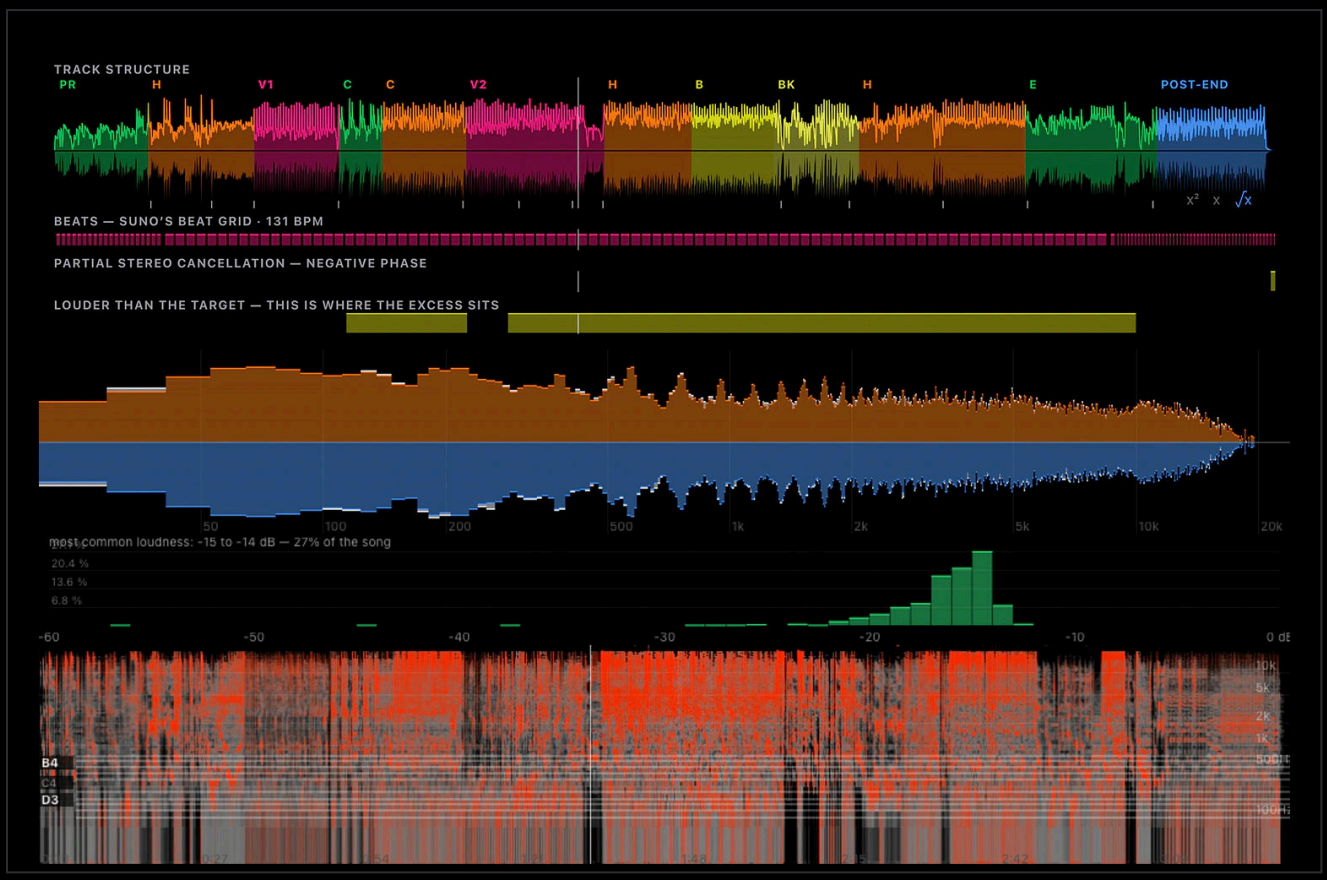


06.1 SOUND SPACE

Your songs as a sky of stars, sorted by sound groups.

Songs that sound alike become neighbours. Size, brightness and corona tell plays, hearts and comments.

- **Closeness is sound**, measured from the audio on your computer, in three dimensions.
- **Fly from song to song** in a ship, a cube, a raven or a Martian saucer.



06.2
ANALYSIS

Every song, measured.

The structure from Suno's sections, the beat grid, and where the level sits too high or the stereo cancels out.

- **The playhead** runs through all lanes, so every finding has its place in the song.
- **Targets for each platform:** streaming, Spotify, Apple Music, YouTube, club, broadcast.

LEVEL AND LOUDNESS	Loudness by EBU R128 · True peak (from the WAV) · Loudness range · Distance to platform targets
FORM	Structure from Suno's sections · Beat grid and BPM · Envelope
STEREO	Phase · Partial cancellation · Width
SOUND	Key · Vocal range · Upper frequency limit · Interfering frequencies
TRACKS	Six instrument tracks · Word timing with Whisper

06.3
WHAT IS MEASURED

Measured at home.

Computed on your computer for every song.

- **No upload:** the audio never leaves the house.

07

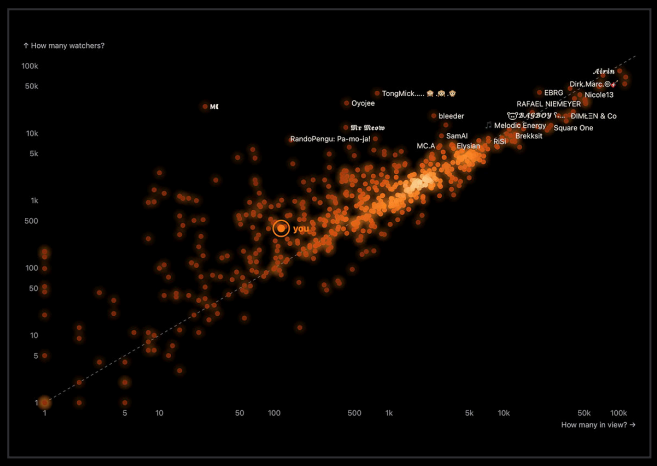
07.1 Hit stream p. 15
07.2 Groupie foam p. 14
07.3 Neighbourhood p. 14
07.4 Community p. 15

LISTENERS

Your music and your listeners.

Which songs move people week by week, and who the people are.





07.2
GROUPIE FOAM

Every listener a cell.

Everyone who gave you hearts or comments, with their avatar. Here sized by loyalty: from first to last reaction.

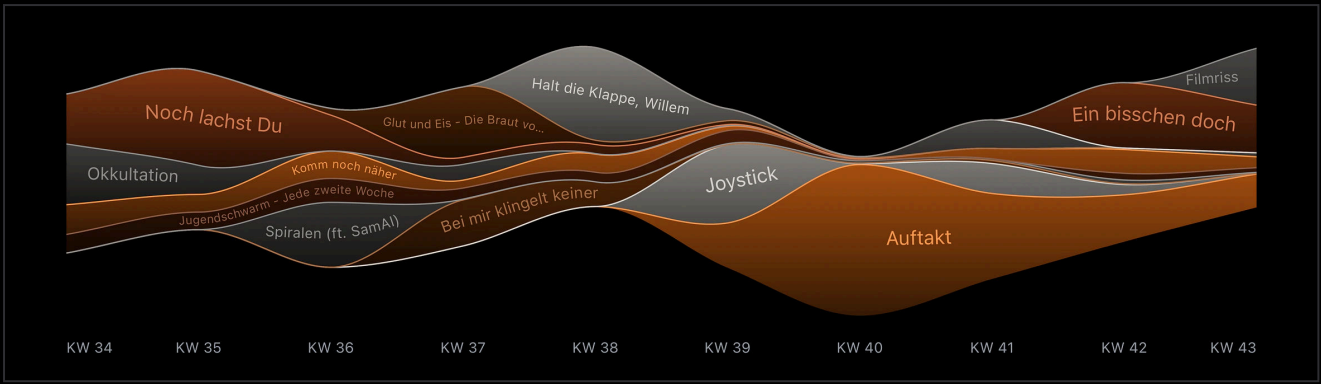
- **Other measures:** hearts, comments, length of comments, recent activity, number of songs.

07.3
NEIGHBOURHOOD

Where you stand.

Everyone around you who makes music: how many they follow and how many follow them. You in orange.

- **Both axes are logarithmic.** On the line, following and followers balance.

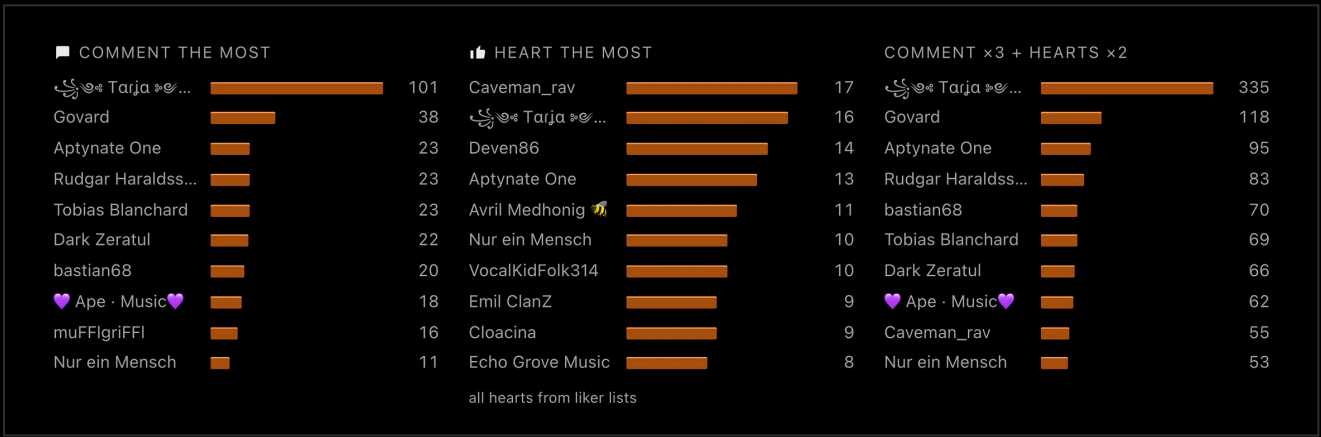


07.1
HIT STREAM

Week by week.

The twelve songs with the most momentum, and their share of the attention each week.

– **Weekly charts** of your own songs below it, by momentum, hearts and plays.



07.4
COMMUNITY

Who is there for you.

Who comments the most, who gives the most hearts, and both together. This month, this year, or all time.

– **A click on a name** opens everything that person wrote to you.



08 YOUR WORK AS A PICTURE

Your whole work. One picture.

As a picture wall with a timeline, or as a foam with a catalogue of works. As a PDF for the print shop.

08.1 Sound foam, picture wall p. 17

08.2 Poster studio p. 17

08.3 Detail p. 18

08.4 Catalogue of works p. 18

08.5 Sketchbook p. 18

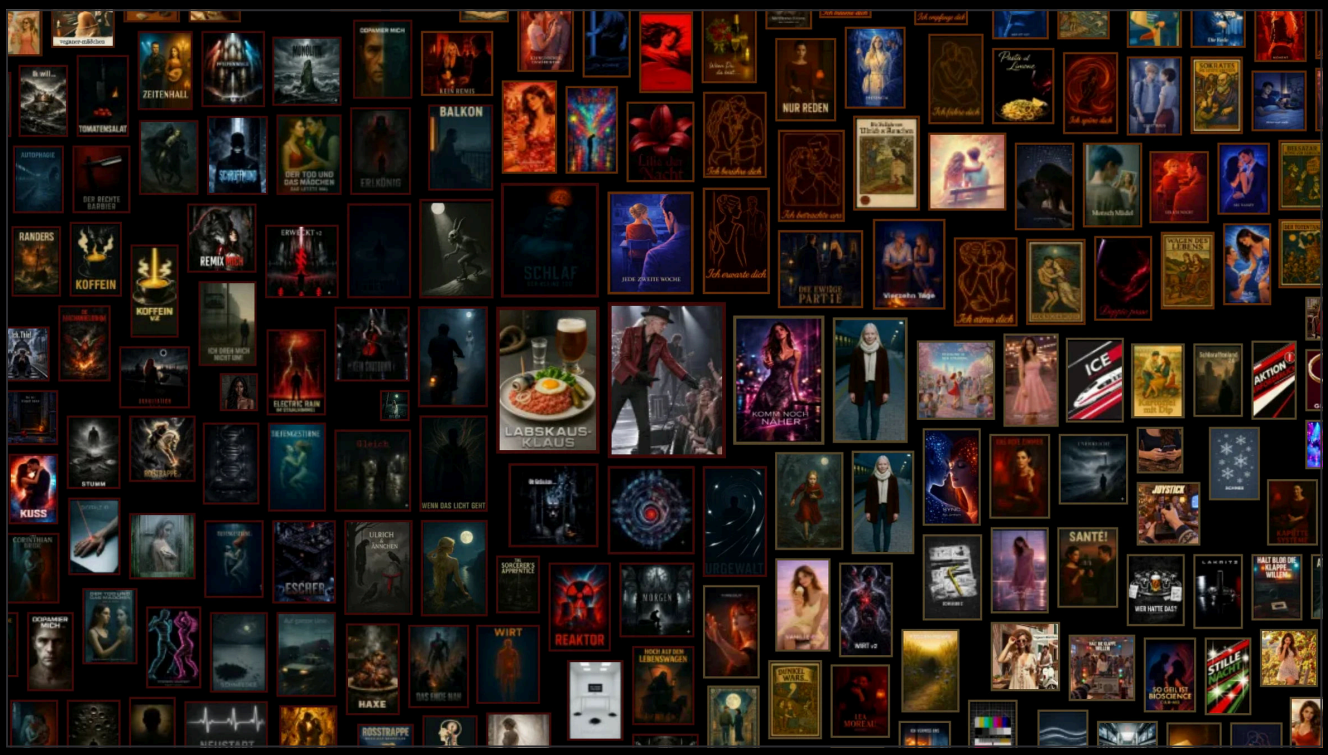
08.6 Four looks p. 18

08.2
POSTER STUDIO
Fig. p. 16

Print it. Hang it.

A PDF export ready for the print shop, from A3 to 100 × 140 cm, free formats up to 3 m. Here the picture show on paper, 100 × 60 cm, with its timeline.

- **Vector text**, images at print resolution, bleed included.
- **Edition and signature line** for a numbered print.



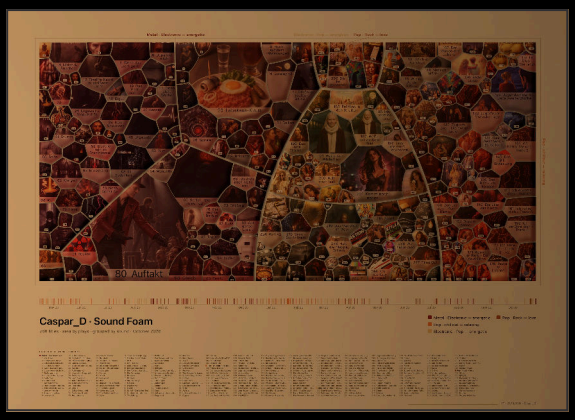
08.1
SOUND FOAM, PICTURE WALL

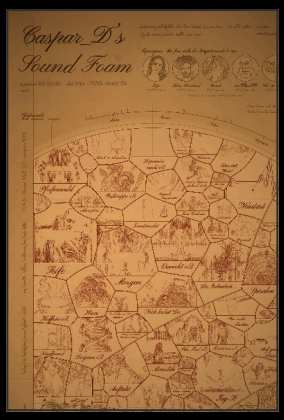
Every song a cell.

Sized by plays, hearts or length. Or the covers hung like a salon exhibition.

- **Grouped by album, genre or space**, coloured by group.
- **Every song keeps its true area.** Zoom in until the smallest cell can be tapped.
- **A tap plays the song.**







08.3
DETAIL

Readable to the last line.

Numbers in the cells, the same numbers in the list.

- **Timeline**: one stroke per song on the day it was made.

08.4
CATALOGUE OF WORKS

Every song with its number.

The sound foam on paper, numbered, the full catalogue below.

- **Grouped by sound**, coloured by group, numbered area by area.

08.5
SKETCHBOOK

In the manner of Leonardo.

The foam as pen drawings on parchment, with seal, studies and notes in the margin.

- **One page of a sketchbook**: title, tabula, instrument studies, refrains.



08.6
FOUR LOOKS

Gallery, paper, stained glass, mosaic.

Every detail adjustable afterwards: ground, joints, glass bands, shadow.

- **Images for social media** in 4:5, 1:1 and story format.

09

CREATE

From cover to clip.

With the effect clip studio, a still picture becomes a moving clip for Suno.

09.1 The picture p. 19
09.2 Depth map p. 19
09.3 Effect clip studio p. 20
09.4 The recipe p. 20





09.1
THE PICTURE

A still cover.

The starting point: the title picture of “Okkultation”.

- **Any picture works:** a cover or one of your own.

09.2
DEPTH MAP

Depth from a flat picture.

The software works out what is near and what is far. Bright is near.

- **Light, fog and focus** sit inside the scene instead of on top of it.



09.3
EFFECT CLIP STUDIO

Bring a cover to life.

Five effects, one of them a camera move, in time with the music. As a loop for Suno.

- **A light mixing desk** drives spotlights, lasers and pulses from the beat.
- **Ken Burns with parallax.**
- **Export a 10-second loop** to upload to Suno as a moving cover.

FLOCK	Sixty dark birds, each dodging its neighbours, all following a wandering centre and respecting the depth map
DUST	240 bright motes rising, laid over the picture as light
COLOUR VEIL	Palette “Ash”, colour dodge at ten percent
COLOUR CHANNEL PULSE	Red, green and blue shift slightly apart on the beat
KEN BURNS MOVE	To the sun, the refinery and across the harbour, with depth of field from the depth map

09.4
THE RECIPE

Five effects.

There are 35 studio effects with 309 settings in all.

- **Saved with the song** and repainted at any time.



10 ON THE SOFA

Lean back with KlangTresor.

On the phone it becomes a music app. On the TV it becomes a stage.

10.1 In the living room p. 22

10.2 Phone p. 22

10.3 TV p. 22



10.1
IN THE LIVING ROOM

Lean back and listen.

The vault runs at home. You listen wherever it is cosy.

- **Any phone in the home network** opens it in the browser, no app to install.



10.2
PHONE

Made for one hand.

Overview, stage, lyrics, Sound Space, foam, details and profile, in portrait or landscape.

- **Everything else waits behind "⋯":** shuffle, repeat, volume, equaliser, details.
- **Sound, effect clip and poster studio** want a big screen and a mouse: any laptop in the home network.

10.3
TV
Fig. p. 21

The stage, big.

Mirror phone, tablet or laptop to the TV, and the lyrics run along.

- **Needs screen mirroring:** AirPlay (from Apple devices) or Google Cast (from Android and Chrome).
- **In landscape the stage fills the screen** and the controls fade away.
- **Karaoke readable from across the room.**

11

STAYS YOURS

A folder. Yours.

Back it up, move it, update the program. The archive stays.

11.1 Export p. 23
11.2 Open source p. 23



11.1 EXPORT

On a stick. For the car too.

Your vault to take along: a frozen archive for playing, without the WAV originals. Double-click its start script and it runs on another Mac or Windows PC. Plus a Music folder: MP3s named by title, cover in the file, for car radio, TV and VLC.

- **In stages:** the program and your newest songs first, instrument tracks last.
- **Stop early** and you still have something to show.

11.2 OPEN SOURCE

Free. MIT.

On GitHub. A bell tells you when an update is out.

- **Updating replaces the program**, never your archive.
- **German, English, Russian.**
- **Third-party parts keep their own licences.** The AI models for musical style are for non-commercial use only; details in web/fremd/LIZENZEN.md.


KlangTresor